

Green Campus Assistant Chatbot using AI

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Abstract:

Developing an AI-powered chatbot designed to promote sustainability and environmental awareness on campus. The chatbot assists students and staff by providing eco-friendly tips, campus event information, waste disposal guidance, and energy-saving practices. It integrates natural language processing (NLP) for seamless interaction and supports a greener, smarter campus.

Problem Statement :

Universities struggle with effectively engaging the campus community in sustainability efforts. Information on green practices is often fragmented or inaccessible, leading to low participation and environmental impact.

Objective :

- To create an AI chatbot that educates and assists campus users in sustainable practices.
- To provide real-time, accurate information on green initiatives and events.
- To automate queries related to recycling, energy use, transportation, and eco-friendly activities.

Data Collection and Preparation :

- Collected data from campus sustainability guidelines, event calendars, waste management manuals, and FAQs.
- Preprocessing involved cleaning and structuring data for training the chatbot (tokenization, stop-word removal, etc.).
- Intent classification and response mapping for training NLP models.

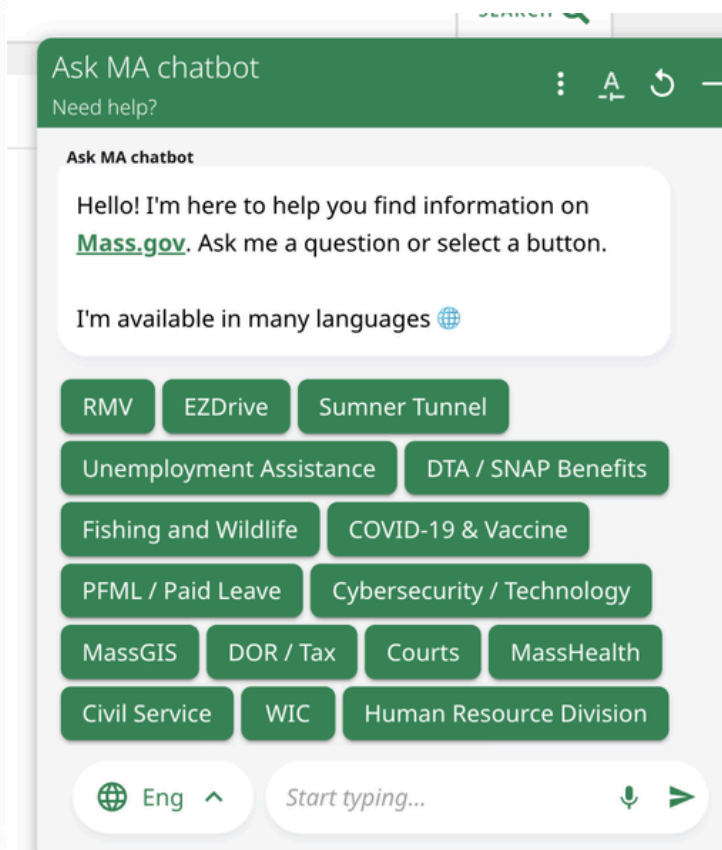
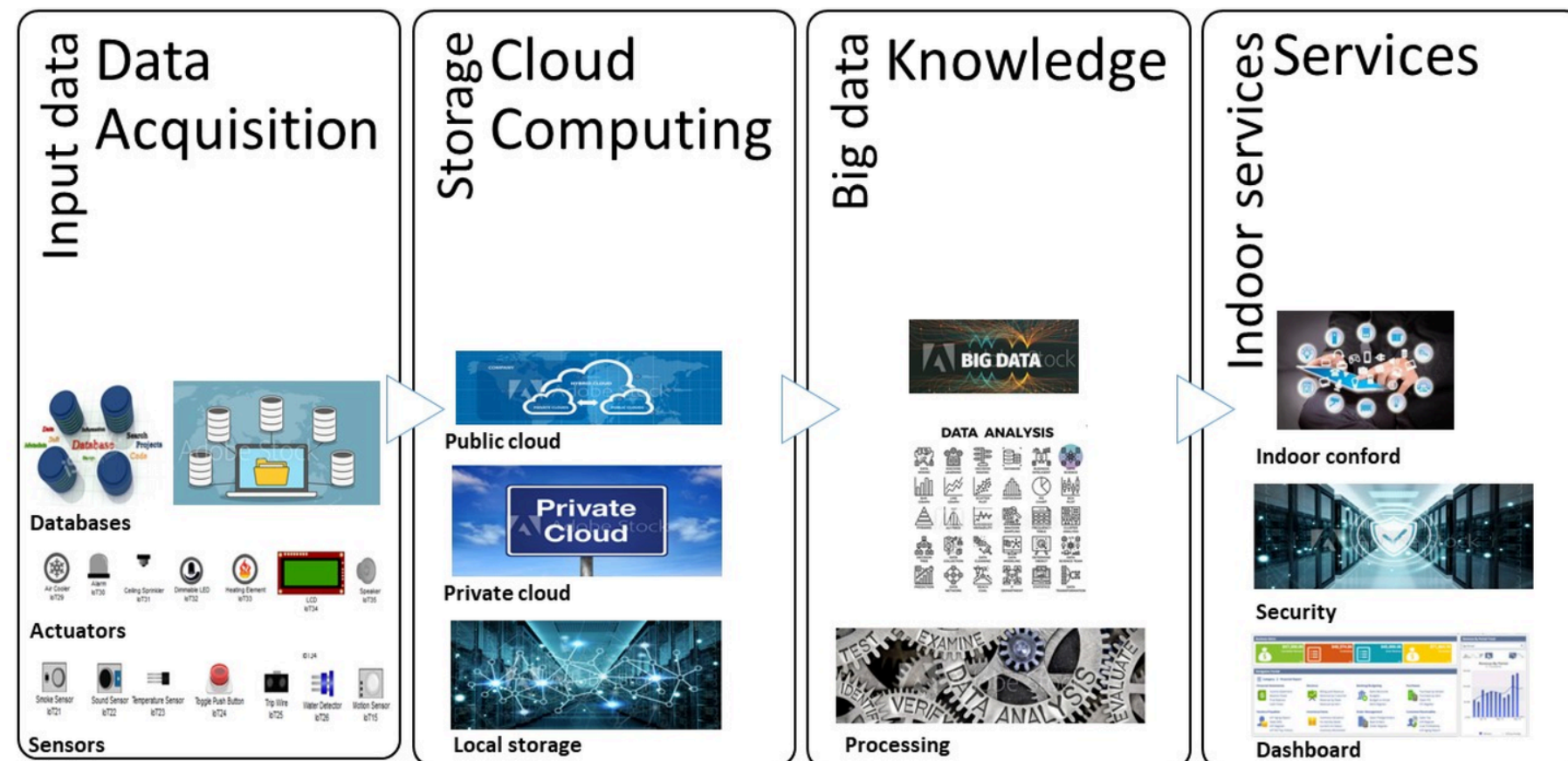
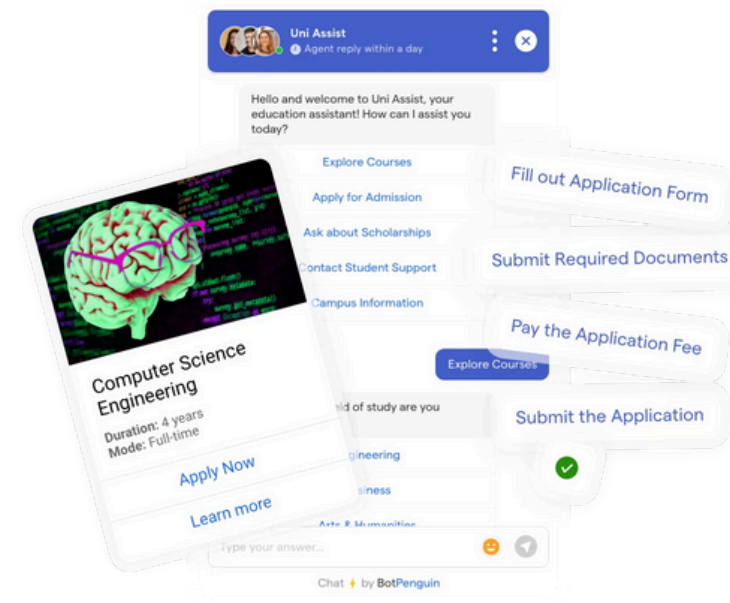
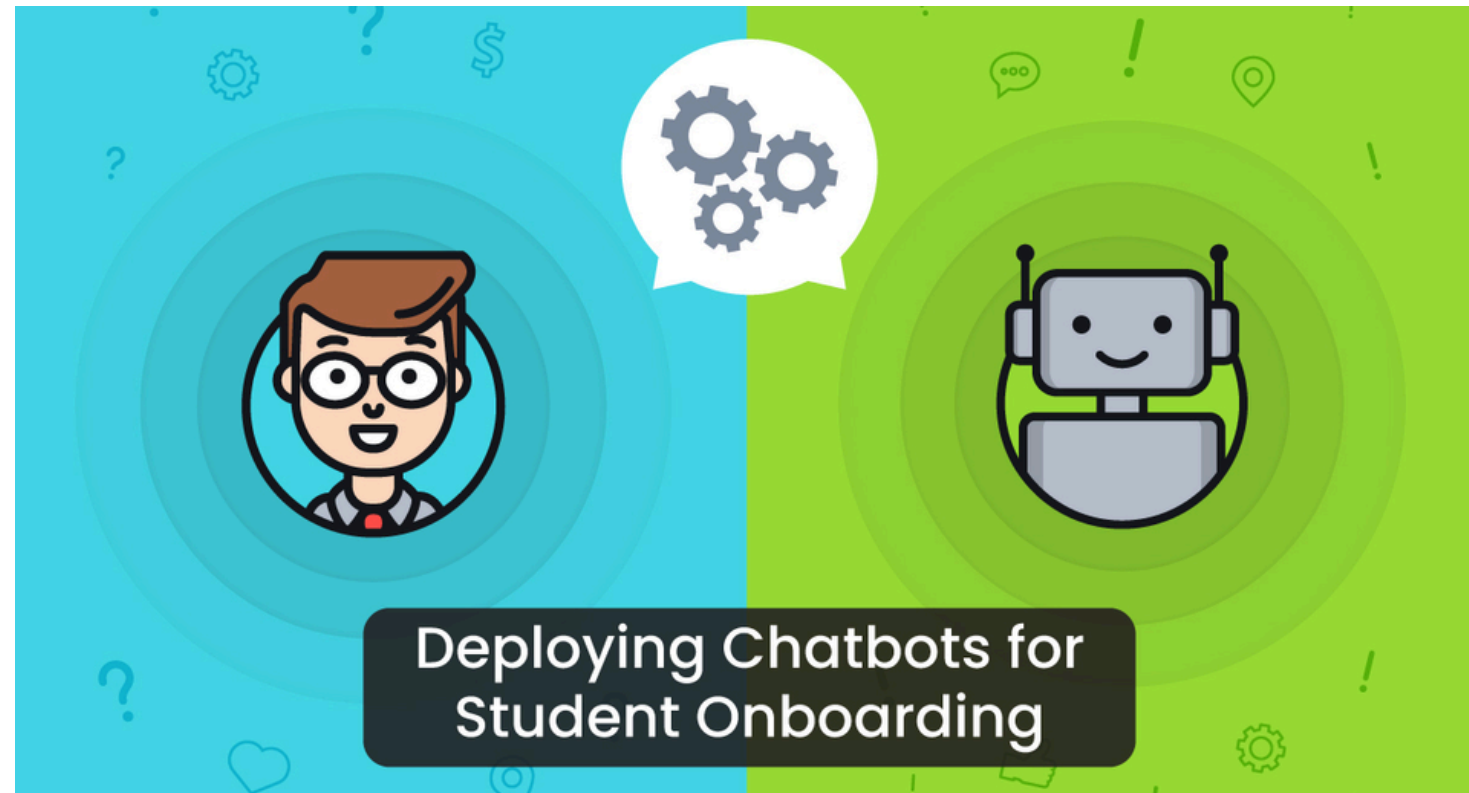
Proposed Solution (Methodology):

- Used a pre-trained NLP model (e.g., BERT or Dialogflow) for conversational AI.
- Integrated with a knowledge base of campus sustainability resources.
- Backend: Python with Flask for API; Frontend: simple web or mobile interface.
- Trained intents like “recycling rules,” “eco events,” “save energy,” etc.

Model Performance Evaluation:

- Accuracy of intent recognition: ~90%
- User satisfaction from pilot testing: positive feedback from 80% of users.
- Confusion matrix and precision/recall metrics used to evaluate performance.

Screenshots / Demonstration (video) :



Future Scope:

- Multilingual support for diverse users.
- Integration with IoT (e.g., smart bins, energy sensors).
- Gamification to increase user engagement.
- AI model improvement with user feedback loop.

Conclusion :

The Green Campus Assistant chatbot is a step toward digital sustainability in education environments. By using AI, it bridges the communication gap and empowers users to contribute to eco-friendly initiatives efficiently and interactively.